Facial Tracking (PTZ)

1. Experiment Objective

The PTZ pan-tilt tracks the face position, keeping the face at the center of the frame.

2. Experiment Principle

1. Key Terms

Term	Description
Visual Tracking	Detects and continuously locks onto a target through a vision algorithm; only the pan-tilt follows, the car does not move
PTZ Pan-Tilt	A two-degree-of-freedom servo pan-tilt; s1 rotates horizontally and s2 tilts, controlled via the `/cmd_ptz_angle` topic
Haar Cascade	An OpenCV cascade classifier based on Haar features that detects face regions using a pretrained XML model
PD Control	Proportional-derivative control; computes the pan-tilt angle step from the target offset (P term) and the offset change rate (D term)
Deadband	When the target offset is within the deadband pixel range, the pan-tilt does not move, avoiding continuous jitter caused by tiny errors
Histogram Equalization	Applies `equalizeHist` to the grayscale image to enhance contrast, improving the Haar detector's face detection rate under varying lighting

2. Technical Architecture

The facial tracking node `face_tracker_ptz_node` subscribes to the raw image topic `/camera/image_raw` of the ESP32 camera, detects face regions with the Haar Cascade classifier, manages the tracking lifecycle with a five-state machine (searching → tracking → lost → paused/identify), computes the pan-tilt angle step with a PD controller, and publishes to `/cmd_ptz_angle` to control the pan-tilt to track the face.

Detection pipeline: image conversion → orientation correction → scaled detection → grayscale conversion → histogram equalization → Haar detectMultiScale → multi-face selection strategy (largest area / nearest center) → target center EMA filter → PD pan-tilt control (fixed 20 Hz).

3. Core Topics Quick Reference

Firmware Uplink (Sensor Data)

Topic	Message Type	QoS	Description
/camera/image_raw	sensor_msgs/msg/Image	best_effort	Raw image from the ESP32 camera

Firmware Downlink (Control Commands)

Topic	Message Type	QoS	Field	Range	Direction
/cmd_ptz_angle	geometry_msgs/msg/Vector3	best_effort + keep_last(1)	x	[-90, 90]°	s1 horizontal servo angle
			y	[-90, 20]°	s2 tilt servo angle

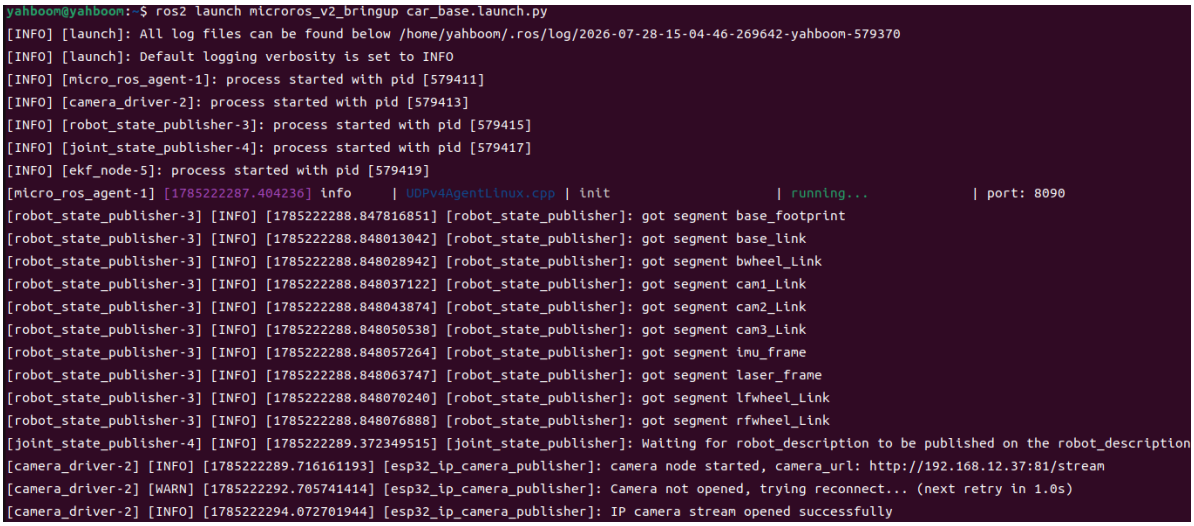
3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑ Supported
No-PTZ version	✗
Required peripherals	ESP32 camera (pan-tilt) + PTZ servo

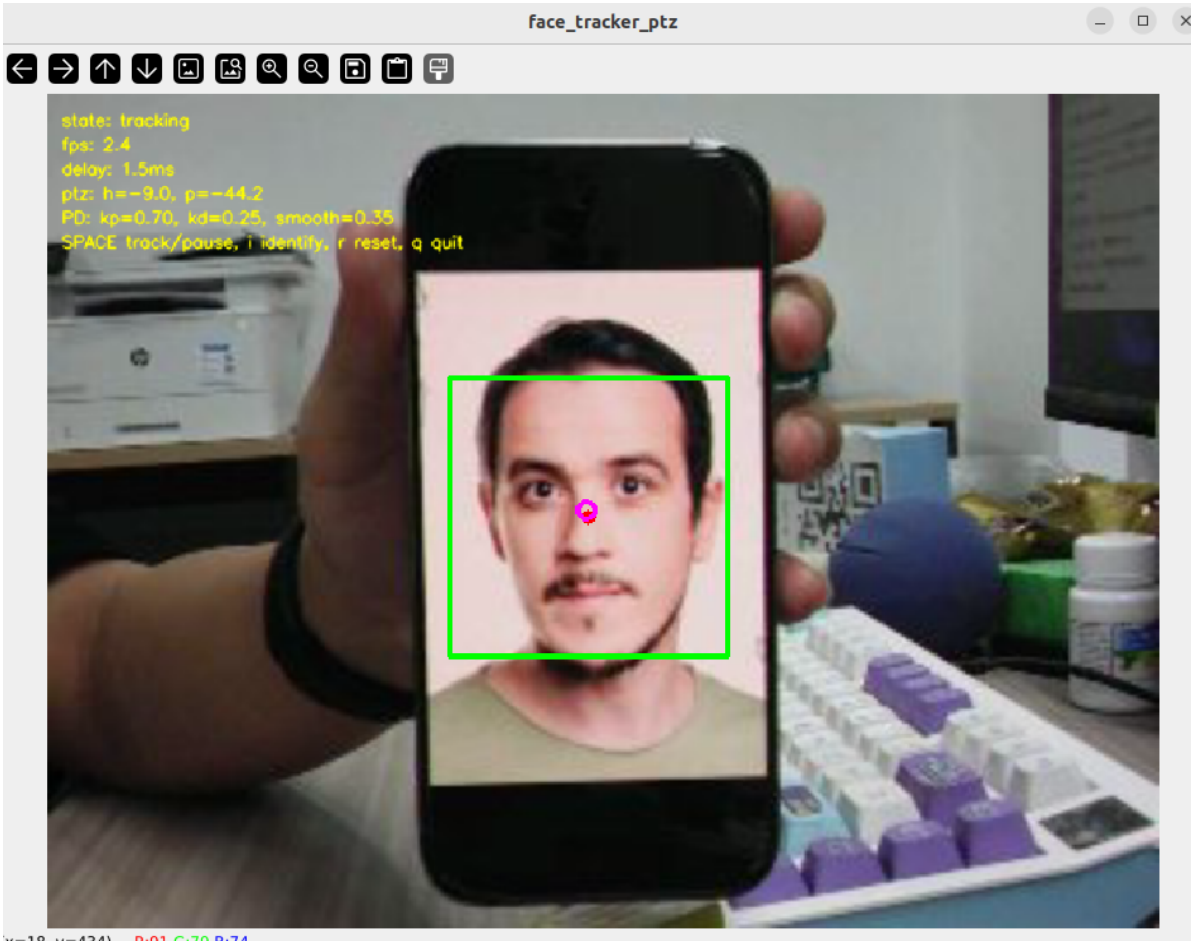
2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```



3. Start Facial Tracking (Terminal 2)

```
ros2 launch microros_v2_ptz_visual_control_pkg face_tracker_ptz.launch.py
```



4. Operation Instructions

Workflow

After startup, the node runs through the main loop `spin_once` plus a fixed-rate image processing timer (`image_processing_rate_hz` = 30 Hz) and an independent pan-tilt control timer (`control_rate_hz` = 20 Hz):

1. Image reception and preprocessing

Subscribes to `/camera/image_raw` (QoS: BEST_EFFORT, KEEP_LAST); the image callback only converts and caches the raw Image. The main loop takes the latest frame at 30 Hz and applies horizontal/vertical flipping and rotation correction as needed.

2. Face detection (Haar Cascade)

Scales the image down by `face_detection_scale` (0.75), converts it to grayscale, optionally applies histogram equalization (`enable_gray_equalize=true`), and detects faces with the OpenCV Haar cascade classifier `detectMultiScale`. Detection parameters: `scaleFactor=1.1`, `minNeighbors=5`, minimum detection size 80×80 px.

3. Multi-face selection

If multiple faces are detected, the main target is chosen by the `face_select_strategy`:

- `largest` (default): selects the face with the largest area
- `nearest_center`: selects the face closest to the frame center

4. Target state management (five-state machine)

The node maintains 5 states: `searching` (searching) → `tracking` (tracking) → `lost` (lost) → can pause to `paused` or switch to `identify` (recognition only). With `auto_start_tracking=true`, the first detected face enters tracking automatically; with `auto_reacquire_face=true`, tracking resumes automatically once a face is re-detected after a loss.

5. Target center low-pass filter

Applies EMA low-pass filtering (α = `target_smooth_alpha` = 0.55) to the detected face center, reducing the effect of Haar detector box jitter on pan-tilt control.

6. PD pan-tilt control

Fixed-rate 20 Hz PD control: normalizes the pixel error (divided by the image half-width/half-height), computes the P+D step, clamps the step (±`max_ptz_step` = 1.2°), smooths the step (α = `ptz_step_smooth_alpha` = 0.35), smooths the angle (α = `ptz_smooth_alpha` = 0.75), and publishes to `/cmd_ptz_angle`.

Interaction

Key	Function	Description
Space	Tracking/pause	Toggles between "tracking" and "paused"; keeps the current pan-tilt angle while paused
i / I	Recognition mode	Only detects the face and draws the box, without controlling the pan-tilt
r / R	Reset target	Clears the current face target and searches again
q / Q / ESC	Exit	Force-exits the node process with `os._exit(0)`

On-screen overlay: The debug window shows the current state (searching/tracking/lost/paused/identify), frame rate (based on the header stamp or reception time), image latency, pan-tilt angle, PD parameters, and key hints in real time. All detected faces are drawn with thin yellow boxes; the primary target face is drawn with a thick green box + red center point, and the filtered target center is a purple hollow circle.

Safety and Edge Cases

- **Classifier load failure**: If the Haar Cascade XML file (`haarcascade_frontalface_default.xml`) fails to load, an error log is recorded and the node does not crash
- **Target lost handling**: When no face is detected for more than `target_lost_timeout_sec` (1.0 s) → switches to the `lost` state; if `lost_target_stop=true`, the filtered target state is cleared and the PD control resets
- **Auto re-acquisition**: With `auto_reacquire_face=true`, a face re-detected after a loss resumes tracking automatically
- **Pan-tilt angle limiting**: Horizontal [-90°, 90°], tilt [-90°, 20°]; initial homing to (0°, -45°)
- **Deadband protection**: The step is forced to zero when the face offset is within the deadband (20 px)
- **Initial publish at startup**: With `publish_initial_ptz_command=true`, the initial pan-tilt angle is published repeatedly 20 times at 0.2 s intervals to ensure the servos receive the homing command

- **Exit strategy**: Press q/ESC to exit directly with `os._exit(0)`; `cv2.destroyAllWindows()` is not called during cleanup to avoid the GUI backend hanging

5. How to Stop

Press `Ctrl+C` to stop Terminal 2 → Terminal 1 in order.

4. Experiment Observations

- **Searching (no face)**: The frame streams in real time with the state `searching`; the Haar classifier keeps scanning and the pan-tilt holds its initial angle (0°, -45°)
- **Face detected + auto tracking**: Enters the `tracking` state automatically; the primary target face is drawn with a thick green box, the face center with a red point, and the filtered tracking point with a purple hollow circle; the pan-tilt follows the face smoothly
- **Multi-face scene**: All detected faces are drawn with thin yellow boxes; the primary target selected by the strategy is highlighted with a thick green box
- **Face within the deadband**: The step is zeroed and the pan-tilt stays still without jitter
- **Face briefly lost (<1 s)**: The last valid face target is retained and tracking is not interrupted
- **Face lost by timeout (>1 s)**: Enters the `lost` state, the PD state resets, and the pan-tilt stays at its last position; with `auto_reacquire_face=true`, tracking resumes automatically when a face is re-detected
- **Paused mode**: The frame continues detecting and showing face boxes, but pan-tilt control is paused and holds the current angle

5. Program Analysis

> Source code paths:
>
> - Launch: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/face_tracker_ptz.launch.py`
> - Node: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/face_tracker_ptz_node.py`
> - Haar classifier: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/config/haarcascade_frontalface_default.xml`

1. Launch Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/face_tracker_ptz.launch.py`
> - Node: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/face_tracker_ptz_node.py`

Parameter	Type	Default Value	Description
`image_topic`	string	`/camera/image_raw`	Raw image input topic
`ptz_command_topic`	string	`/cmd_ptz_angle`	Pan-tilt angle output topic
`show_debug_window`	bool	`true`	Whether to show the OpenCV debug window
`window_name`	string	`face_tracker_ptz`	OpenCV debug window name
`face_cascade_file`	string	`config/haarcascade_frontalface_default.xml`	Haar face cascade classifier path
`face_detection_scale_factor`	float	`1.1`	Haar detectMultiScale scale factor
`face_min_neighbors`	int	`5`	Haar detectMultiScale minimum neighbors
`face_min_width` / `face_min_height`	int	`80` / `80` px	Minimum face detection size
`face_detection_scale`	float	`0.75`	Image scale factor before face detection (0.1~1.0)
`face_select_strategy`	string	`largest`	Multi-face selection strategy: largest / nearest_center
`enable_gray_equalize`	bool	`true`	Whether to apply histogram equalization to the grayscale image
`pd_kp` / `pd_kd`	float	`0.7` / `0.25`	PD proportional/derivative gains
`control_rate_hz`	float	`20.0` Hz	Pan-tilt control publishing rate
`image_processing_rate_hz`	float	`30.0` Hz	Image processing main-loop rate
`deadband_x` / `deadband_y`	float	`20.0` / `20.0` px	Horizontal/vertical deadband
`target_smooth_alpha`	float	`0.55`	Face center low-pass filter coefficient
`ptz_smooth_alpha`	float	`0.75`	Pan-tilt angle low-pass filter coefficient
`max_ptz_step`	float	`1.2`°	Maximum single angle step
`ptz_step_smooth_alpha`	float	`0.35`	Step smoothing coefficient
`target_lost_timeout_sec`	float	`1.0` s	Target lost timeout
`auto_start_tracking`	bool	`true`	Start tracking automatically after a face is detected
`auto_reacquire_face`	bool	`true`	Resume automatically when a face is re-detected after a loss
`lost_target_stop`	bool	`true`	Whether to clear the filtered state after the target is lost
`initial_horizontal_deg` / `initial_pitch_deg`	float	`0.0`° / `-45.0`°	Initial pan-tilt angles
`horizontal_min_deg` / `horizontal_max_deg`	float	`-90.0`° / `90.0`°	s1 horizontal servo angle range
`pitch_min_deg` / `pitch_max_deg`	float	`-90.0`° / `20.0`°	s2 tilt servo angle range

2. Core Node Analysis

`FaceTrackerPtzNode` inherits from `rclpy.node.Node` and is the core implementation of facial tracking. The constructor, the pan-tilt control callback, and the PD computation method are shown below.

Constructor `__init__`

The constructor declares the ROS2 parameters, loads the Haar classifier, creates the publishers and subscribers, initializes the tracking state, and sets up the control timer:

```
class FaceTrackerPtzNode(Node):
    """ROS2 face tracking PTZ control node."""

    def __init__(self) -> None:
        super().__init__("face_tracker_ptz_node")

        self.latest_image_lock = threading.Lock()
        self.target_state_lock = threading.Lock()

        self.latest_image_frame = None
        self.tracking_state = "searching"

        self._declare_ros_parameters()
        self._load_ros_parameters()
        self._reload_face_cascade_classifier()

        # Initial pan-tilt angles
        self.horizontal_angle_deg = self.initial_horizontal_deg
        self.pitch_angle_deg = self.initial_pitch_deg

        # Create the image subscriber (raw image)
        self.image_subscription = self.create_subscription(
            Image,
            self.image_topic_name,
            self.image_callback,
            QoSProfile(history=QoSHistoryPolicy.KEEP_LAST, depth=1,
                       reliability=QoSReliabilityPolicy.BEST_EFFORT,
                       durability=QoSDurabilityPolicy.VOLATILE),
        )

        # Create the pan-tilt angle publisher
        self.ptz_angle_publisher = self.create_publisher(
            Vector3, self.ptz_command_topic_name, 10
        )

        # Create the pan-tilt control timer (fixed rate, 20 Hz by default)
        self._recreate_ptz_control_timer(False)

        # Register the runtime dynamic tuning callback
        self.dynamic_parameter_callback_handle =
self.add_on_set_parameters_callback(
            self.on_dynamic_parameter_update
        )

        # Initialize the OpenCV debug window
        self._initialize_debug_window()
```

```
        # Start repeated publishing of the initial pan-tilt angle
        if self.publish_initial_ptz_command:
            self._start_initial_ptz_publish_timer()
```

Pan-tilt control callback `ptz_control_timer_callback`

Triggered at a fixed rate (20 Hz); reads the target snapshot and runs PD control only in the `tracking` state:

```
def ptz_control_timer_callback(self) -> None:
    if self.is_shutdown_requested:
        return

    current_time_sec = time.monotonic()
    time_delta_sec = current_time_sec - self.last_control_time_sec
    self.last_control_time_sec = current_time_sec

    if self.tracking_state != "tracking":
        self._reset_pd_control_state()
        return

    target_state = self._get_face_target_state_snapshot()
    if not self._is_target_snapshot_usable(target_state):
        self._handle_unusable_target_snapshot(target_state)
        return

    image_cx = float(target_state.image_width) * 0.5
    image_cy = float(target_state.image_height) * 0.5

    horizontal_error = float(target_state.filtered_target_center_x) - image_cx
    pitch_error = float(target_state.filtered_target_center_y) - image_cy

    self.calculate_ptz_angle_command(
        horizontal_error, pitch_error, time_delta_sec
    )
    self.publish_ptz_angle_command()
```

PD control core `calculate_ptz_angle_command`

Implements the complete PD control chain from the face offset to the pan-tilt angle step:

```
def calculate_ptz_angle_command(
    self, horizontal_error_pixel, pitch_error_pixel, time_delta_sec
) -> None:
    # Deadband handling
    if abs(horizontal_error_pixel) <= self.deadband_x:
        horizontal_error_pixel = 0.0
    if abs(pitch_error_pixel) <= self.deadband_y:
        pitch_error_pixel = 0.0

    # Normalize the pixel error
    image_cx = float(self.latest_tracking_image_width) * 0.5
    image_cy = float(self.latest_tracking_image_height) * 0.5
    norm_x = horizontal_error_pixel / image_cx
    norm_y = pitch_error_pixel / image_cy
    time_delta_sec = max(float(time_delta_sec), 1.0e-3)
```

```
    # Derivative terms
    dx = (norm_x - self.previous_error_x) / time_delta_sec
    dy = (norm_y - self.previous_error_y) / time_delta_sec

    # PD step computation + limiting
    raw_x = (self.pd_kp * norm_x + self.pd_kd * dx) * self.max_ptz_step
    raw_y = (self.pd_kp * norm_y + self.pd_kd * dy) * self.max_ptz_step
    raw_x = clamp_numeric_value(raw_x, -self.max_ptz_step, self.max_ptz_step)
    raw_y = clamp_numeric_value(raw_y, -self.max_ptz_step, self.max_ptz_step)

    # Step EMA smoothing
    alpha = self.ptz_step_smooth_alpha
    step_x = (1.0 - alpha) * self.last_step_x + alpha * raw_x
    step_y = (1.0 - alpha) * self.last_step_y + alpha * raw_y

    # Save the control history
    self.previous_error_x = norm_x
    self.previous_error_y = norm_y
    self.last_step_x = step_x
    self.last_step_y = step_y

    # Direction reversal
    if self.reverse_horizontal_direction:
        step_x = -step_x
    if self.reverse_pitch_direction:
        step_y = -step_y

    # Compute the desired angle and clamp it
    desired_h = self.horizontal_angle_deg + step_x
    desired_p = self.pitch_angle_deg + step_y
    desired_h = clamp_numeric_value(desired_h, self.horizontal_min_deg,
                                    self.horizontal_max_deg)
    desired_p = clamp_numeric_value(desired_p, self.pitch_min_deg,
                                    self.pitch_max_deg)

    # Angle EMA smoothing
    ptz_alpha = self.ptz_smooth_alpha
    self.horizontal_angle_deg = (
        (1.0 - ptz_alpha) * self.horizontal_angle_deg + ptz_alpha * desired_h
    )
    self.pitch_angle_deg = (
        (1.0 - ptz_alpha) * self.pitch_angle_deg + ptz_alpha * desired_p
    )

    # Final limiting
    self.horizontal_angle_deg = clamp_numeric_value(
        self.horizontal_angle_deg, self.horizontal_min_deg,
        self.horizontal_max_deg)
    self.pitch_angle_deg = clamp_numeric_value(
        self.pitch_angle_deg, self.pitch_min_deg, self.pitch_max_deg)
```

6. Frequently Asked Questions

Problem	Cause	Solution
Profile tracking unstable	Haar Cascade is sensitive to frontal faces; profiles lack enough features	Face the camera squarely, or reduce strictness by adjusting `face_detection_scale_factor`
Face not detected	Insufficient lighting or too far away	Improve the lighting, move closer to the camera, and make sure `enable_gray_equalize=true`
Wrong face tracked with multiple faces	The largest face is selected by default; the nearest-center strategy may suit better	`ros2 param set /face_follow_node face_select_strategy nearest_center`
Pan-tilt responds slowly	`ptz_smooth_alpha` too large causes over-smoothing	Reduce `ptz_smooth_alpha` (e.g., 0.5) to improve responsiveness